# Monodip Bhattacharya

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## Education

**Dr. BC Roy Engineering College, Durgapur** — Durgapur, India
*B.Tech in Electronics and Communication Engineering* — *Aug 2022 – May 2026*
- CGPA: 6.5/10

**Indira Gandhi Memorial High School** — Kolkata, India
*Class XII – CBSE* — *May 2020 – March 2022*

**St. Stephen's School** — Kolkata, India
*Class X – ICSE* — *May 2019 – March 2020*

## Technical Skills

**Languages:** JavaScript (ES6+), Python, TypeScript, HTML5, CSS3, SQL
**Frameworks & Libraries:** React.js, Node.js, Express.js, Vite, PDF.js, Puter.js, Socket.io, TailwindCSS, DaisyUI
**Databases:** MongoDB, MySQL
**Tools & Platforms:** Git, GitHub, REST APIs, JWT, bcrypt, Cloudinary, Arduino, IoT Protocols
**Concepts:** Data Structures, Full-Stack Development, Real-Time Systems, Firmware Development, Encryption (PBKDF2, Fernet), CRUD Operations

## Work Experience

**Embedded Systems & IoT Intern** — June 2025 – July 2025
*IC Design and Fabrication Center, Dept. of ECE, Jadavpur University* — *Kolkata, India*
- Programmed Arduino microcontrollers to interface with sensors and transmit data over IoT protocols, building end-to-end hardware-software pipelines covering circuit design, firmware development, and IoT data flow architecture.
- Delivered a complete real-time IoT project within a 2-month timeline; received official certification upon project completion.

## Projects

**Chatify** | *React.js, Node.js, Express, MongoDB, Socket.io, TailwindCSS, Cloudinary, Zustand* | GitHub | Live — 2026
- Built a production-ready real-time chat application with Socket.io and Node.js, handling bidirectional data synchronization and live messaging across concurrent users.
- Engineered a scalable MongoDB/Mongoose database layer with cursor-based pagination and infinite scrolling to optimize query performance and reduce payload sizes.
- Secured authentication with JWT stored in HTTP-only cookies to prevent XSS attacks; implemented read receipts, live typing indicators, and message editing/deletion.
- Integrated Cloudinary for scalable media hosting and built a fully responsive UI with 30+ switchable themes using TailwindCSS and DaisyUI.

**AI Resume Analyzer** | *React, Vite, PDF.js, Puter.js AI, REST APIs* | GitHub | Live — 2025
- Built an AI-powered resume analysis tool using React, Vite, and PDF.js that parses uploaded PDFs and delivers ATS compatibility scores, grammar feedback, keyword suggestions, and recommended job roles via a responsive analytics dashboard powered by Puter.js AI.

**AI Thumbnail Generator (Thumbfy)** | *MERN Stack, JWT, bcrypt, REST APIs* | GitHub — 2025
- Architected a full-stack AI thumbnail generation platform using the MERN stack, serving end-to-end image creation workflows from a responsive React frontend with a clean, intuitive user interface.
- Implemented JWT-based authentication and bcrypt password hashing via RESTful Express.js APIs backed by MongoDB, enabling secure per-user AI generation sessions.

**Encrypted Password Manager** | *Python, PBKDF2, Fernet Encryption, JSON* | GitHub — 2024
- Engineered a CLI password manager in Python using PBKDF2 key derivation and Fernet symmetric encryption, with master-password-gated access and encrypted JSON storage ensuring zero plaintext credential exposure.

## Additional

**Interests:** Football, Music, Reading, Travelling
**Soft Skills:** Problem-solving, Analytical Thinking, Time Management, Team Collaboration, Communication